Parental Incarceration and Children's Educational Outcomes

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Postdoc Feedback Group TT2026

What do I need?

- Old paper (MPhil thesis) drafted but never submitted
- Thinking of revising and submitting to ***Sociology of Education***
- Does the **framing** work? Blind spots? More outcomes? **What to exclude**?
- I'll show loads of findings — want your **reactions**

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- **RQ:** Does parental incarceration causally impact educational outcomes?

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Intuition: "Lucky" vs "unlucky" sibling

$N \approx 890,159$ — 16 full birth cohorts
$N(id) \approx 8,000$ – $28,000$

Prior literature

1. **Descriptive**: large negative associations $\approx$ 15–22 pp in HS completion (Andersen 2016; Hagan and Foster 2012; Cho 2011)

2. **Matching**: Effects shrink substantially with observed controls; some to zero (Haskins 2014, 2016; Turney 2017; Turney and Haskins 2014)

3. **Judge IV:** mixed evidence — negative in SE, null in NO, *positive* in US and CO (Bhuller et al. 2018; Dobbie et al. 2018; Norries et al. 2021; Arteaga 2023)

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Possible contributions:

- IVs identify very narrow LATEs (marginal defendants)
- Intensive versus extensive margins
- Gender heterogeneity
- Mechanisms (discuss how later)

Theoretical pathways

1. **Stigma / labelling** (Goffman; Link & Phelan)
   - Associative stigma — “a prisoner’s child” becomes a discrediting attribute.
   - Operates externally (teachers, peers) and internally (self-perception, secondary deviance).
   - ***Prediction***: *prevalence matters, dosage less so.*

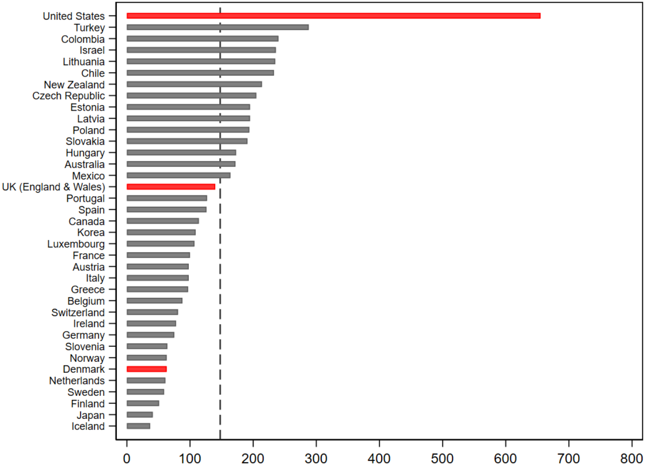
2. **Stress / strain** (Agnew; McLanahan)
   - Parental absence + economic loss + family instability.
   - ***Prediction***: *dosage should matter; longer/more spells worse.*

3. **Human capital / social learning** (Cunha & Heckman; Hagan & Dinovitzer)
   - Disrupted skill formation; bad role models; degraded home learning environment.
   - ***Prediction***: *earlier exposure worse; same-sex parent loss especially damaging.*

I don’t formally test mechanisms — but the timing, dosage, and gendered analyses speak to them.

Why Denmark

Figure: Prison Population Per 100,000 in OECD Member Countries, 2018.



Why Denmark?

A “least-likely” case:

- Incarceration rate $\approx$ 63 per 100,000 (vs. 655 in US).
- Short sentences (median 10 months; 17% over 12 months).
- Extensive welfare state, no tuition fees, universal benefits.
- Expanded use of community service & electronic monitoring since 1990s.

If effects exist here, they likely exist elsewhere.

Also:

- Registers cover the universe of residents.
- Linkable parent–child–sibling IDs.
- Exact prison admission/release dates.
- Educational records with completion and enrollment dates.

The data make a sibling FE design with $\sim$28K identifying observations feasible. Nothing comparable exists in survey data.

Empirical strategy

Baseline sibling FE model:

$$y_{ij} = \theta D_{ij} + \mathbf{x}_{ij}'\boldsymbol{\beta} + \lambda t_{ij} + \alpha_j + \varepsilon_{ij}$$

- y_{ij}: HS completion by age 20 for child i in family j.
- D_{ij}: 1 if child experienced any parental incarceration before age 15.
- α_j: family fixed effect — absorbs all unobserved characteristics shared by siblings.
- $\mathbf{x}_{ij}$: sibling-varying controls (sex, birth weight, APGAR).

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Identification:

- Effective sample = families where at least one sibling treated, one not.
- $\sim$28K identifying obs. (mother FE) / $\sim$8K (family FE).
- Two key assumptions: (i) family environment is sibling-invariant; (ii) parental incarceration not a response to child's endowments.
- I report **mother FE** (shares mom, possibly different dads) and **family FE** (shares both parents) separately.

Sample: descriptive gap

	Parent incarcerated ($n = 52{,}417$)	Parent not incarcerated ($n = 837{,}742$)
HS completion by age 20	0.20	0.42
Birth weight (kg)	3.26	3.42
Non-Western immigrant (father)	0.09	0.05
Mother: compulsory school only	0.66	0.33
Father: compulsory school only	0.55	0.25
Mother employed at birth	0.49	0.75
Father employed at birth	0.60	0.88

- **Raw 22pp gap in HS completion.**
- Families differ on essentially every observable.

Main result: estimates shrink as design tightens

	Pooled OLS		Mother FE	Family FE
	No controls	Full controls	+ controls	+ controls
Parent incarcerated	-0.223^{***}	-0.092^{***}	-0.038^{***}	-0.016
	(0.002)	(0.002)	(0.005)	(0.008)
N (identifying)	890,159	890,159	28,328	8,264

- 22pp raw $\rightarrow$ 9pp with observed controls $\rightarrow$ 4pp with mother FE $\rightarrow$ 1.6pp (n.s.) with family FE.
- **Most of the descriptive gap is selection**
- Family FE estimates are smaller *and* less precise.

Mother FE vs family FE (full siblings)

Why estimates differ: in mother FE, comparison may involve siblings whose *family environment* differs (because the father changed). The fixed effect doesn't actually hold environment constant.

Test: interact treatment with "children have different fathers" indicator in the mother sample.

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	Estimate	**SE**
Parent incarcerated (same father)	-0.004	(0.009)
Different father	-0.033^{***}	(0.008)
Parent incarcerated $\times$ Diff. father	-0.033^{**}	(0.010)

- The mother FE "effect" is concentrated where the father changes.
- This is consistent with family environment shifts, not with parental incarceration per se.
- **Methodological contribution:** mother-ID sibling FE designs are biased in populations where fathers change.

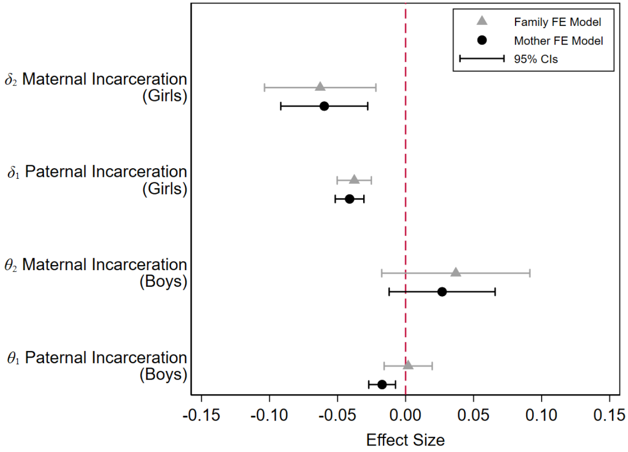
Gender heterogeneity: gendered loss

Same but interact gender of incarcerated parent (D^P, D^M) with gender of child (F_{ij}):

$$y_{ij} = \gamma F_{ij} + \theta_1 D^P_{ij} + \theta_2 D^M_{ij} + \delta_1 (F_{ij} \times D^P_{ij}) + \delta_2 (F_{ij} \times D^M_{ij}) ... \alpha_j$$

Gender heterogeneity: gendered loss

- **Girls more harmed by parental incarceration** (internalizing behaviors?)
- Same-sex parent-child pairing is worst (gendered loss)



Timing effects

Model: dummies for age at first parental incarceration (1–6, 7–12, 13–16, 17–20). Reference: never.

First parental incarceration at age...	Mother FE	Family FE
1–6	-0.040^{***}	-0.022^{*}
7–12	-0.047^{***}	-0.034^{**}
13–16	-0.038^{***}	-0.025
17–20	-0.018	-0.010

- Effects peak in **ages 7–12** — primary school.
- No detectable effect when incarceration occurs after age 17.
- Consistent with:

  Stigma activated by school context (children are old enough to be known).
  Skill formation disruption at a sensitive developmental window.

No dosage effects

Duration (less than 3mo / 3–12mo / 12mo+):

	Mother FE
$<$3 months	-0.041^{***}
3–12 months	-0.039^{***}
$>$12 months	-0.031^{***}

Frequency (1, 2, 3+ spells):

	Mother FE
1 spell	-0.033^{***}
2 spells	-0.057^{***}
3+ spells	-0.043^{***}

- **Prevalence matters; dosage doesn't.** One short spell is about as harmful as many long ones.
- Harder to square with **strain** (which predicts accumulation) than with **stigma** (which activates on disclosure).

What I think the paper shows

1. **The descriptive gap is mostly selection.** 22pp $\rightarrow$ 4pp (or smaller) when family-level unobservables are absorbed.
2. **There *is* a causal effect.** Modest in magnitude, but consistent across specifications and roughly aligned with recent quasi-experimental Nordic work.
3. **Effects are gendered along same-sex lines.** Most pronounced for daughters of incarcerated mothers.
4. **Effects peak at primary-school age (7–12).** Prevalence, not dosage, drives them.
5. **Methodological note:** maternal-ID sibling FE designs are biased in populations where mothers change partners.

Implicit theoretical implication: consistent with stigma + sensitive-period skill formation; harder to square with pure strain (but makes sense in DK?).

What to do?

1. **Headline:** What is the main story? Selection? Family changes? Gendered loss?

2. **Add more educational outcomes?**

3. **Teacher evaluations vs written exams (stigma?)**

4. **Target journal** *Sociology of education? Criminology? Social Forces?*